

[● Complete Academic & Exam Guide](#)

Cellular Respiration in Plant Physiology

An integrated biochemical analysis of energy extraction from respiratory substrates, spanning Glycolysis in the cytoplasm to Oxidative Phosphorylation inside mitochondrial cristae. Fully compliant with NCERT Class 11 standards and NEET high-yield requirements.

OVERALL BALANCED CHEMICAL EQUATION



Cellular Sub-Compartment Map

[Interactive](#)

1. Cytoplasm

Glycolysis (EMP Pathway)

Anaerobic • Yields 2 Pyruvate

2. Mitochondrial Matrix

Link Rxn & Krebs Cycle

Aerobic • Full CO₂ Decarboxylation

3. Inner Membrane

ETS (Complexes I - IV)

Oxidizes NADH & FADH₂

4. F₀-F₁ Particle

ATP Synthase (Complex V)

4 H⁺ per 1 ATP Synthesized

12.1 Do Plants Breathe?

[NCERT Pages 154–155](#)

Plants require oxygen (O₂) for respiration and release carbon dioxide (CO₂), but they do not possess specialized respiratory organs such as lungs or gills. Gaseous exchange occurs primarily via **stomata** (in leaves and green stems) and **lenticels** (in woody stems and roots).

3 FUNDAMENTAL REASONS WHY PLANTS MANAGE WITHOUT RESPIRATORY ORGANS:

1. **Independent Gas Exchange:** Each plant part (root, stem, leaf) takes care of its own gaseous exchange needs with minimal inter-part transport required.
2. **Low Volumetric Demand:** Plants do not demand large volumes of gas exchange. Roots, stems, and leaves respire at far lower rates than animals. Photosynthesis produces abundant internal O₂ directly within green cells.
3. **Short Diffusion Distance:** Living cells are located close to the surface. Even bulky stems have living cells arranged in thin layers beneath bark (with lenticels), while interior cells are dead structural tissues. Loose parenchyma packing creates interconnected air spaces.

TABLE 12.1: CELLULAR RESPIRATION VS COMBUSTION

FEATURE	CELLULAR RESPIRATION	COMBUSTION
Location	Inside living cells (Cytoplasm & Mitochondria)	Non-cellular physical/chemical process
Control	Step-wise enzyme-controlled biological process	Uncontrolled high-temperature physical burning
Energy Output	>40% trapped in chemical bonds as ATP	100% liberated instantly as heat and light

12.2 Glycolysis (EMP Pathway)

NCERT Pages 155–156

Discovered by Gustav Embden, Otto Meyerhof, and J. Parnas • Universal Cytoplasmic Pathway

Glycolysis originates from Greek '*Glycos*' (sugar) + '*Lysis*' (splitting). It occurs in the **cytoplasm** of all living organisms and requires no molecular oxygen (O₂).

In plant cells, glucose is derived from **sucrose** (the end product of photosynthesis) or storage carbohydrates. Sucrose is converted into glucose and fructose by the enzyme **Invertase**.

💡 **MEMORY RULE FOR GLYCOLYSIS ATPs**
"Pay 2 ATPs early (Steps 1 & 3), collect 4 ATPs + 2 NADH late (Steps 6, 7, 10)!"

Net Gain: 2 Pyruvic Acid + 2 ATP + 2 NADH + 2 H⁺

TABLE 12.2: COMPLETE 10-STEP REACTION SEQUENCE OF GLYCOLYSIS

STEP	REACTION TRANSFORMATION	ENZYME	ENERGETICS
Step 1	Glucose + ATP → Glucose-6-Phosphate + ADP	Hexokinase	-1 ATP
Step 2	Glucose-6-P ⇌ Fructose-6-P	Phosphohexose Isomerase	Isomerisation
Step 3	Fructose-6-P + ATP → Fructose-1,6-Bisphosphate + ADP	Phosphofructokinase (PFK)	-1 ATP
Step 4	Fructose-1,6-BP ⇌ PGAL (3C) + DHAP (3C)	Aldolase	Cleavage into 2×3C
Step 5	DHAP ⇌ Glyceraldehyde-3-Phosphate (PGAL)	Triose Phosphate Isomerase	Yields 2x PGAL
Step 6	2 PGAL + 2 H ₃ PO ₄ + 2 NAD ⁺ → 2 BPGA + 2 NADH + 2 H ⁺	PGAL Dehydrogenase	+2 NADH
Step 7	2 1,3-BPGA + 2 ADP → 2 3-PGA + 2 ATP	Phosphoglycerate Kinase	+2 ATP (SLP)
Step 8	2 3-PGA ⇌ 2 2-Phosphoglycerate (2-PGA)	Phosphoglyceromutase	Shift
Step 9	2 2-PGA → 2 Phosphoenolpyruvate (PEP) + 2 H ₂ O	Enolase (Mg ²⁺)	Dehydration

12.3 Fermentation (Anaerobic Pathways)

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1. ALCOHOLIC FERMENTATION

e.g., Yeast

Pyruvic acid is converted into Ethanol and CO₂ in two enzymatic steps:

1. Pyruvic Acid → Acetaldehyde + CO₂
(Pyruvic acid decarboxylase)
2. Acetaldehyde + NADH + H⁺ → Ethanol + NAD⁺
(Alcohol dehydrogenase)

⚠ **Critical NCERT Threshold:** Yeast poisons itself to death when alcohol concentration reaches **~13%**. Higher proof beverages require distillation.

2. LACTIC ACID FERMENTATION

e.g., Muscle / LAB

Occurs in fatigued animal muscle cells during oxygen deficiency or in Lactic Acid Bacteria:



(Enzyme: Lactate Dehydrogenase)

⚡ **Note:** No CO₂ is evolved in Lactic Acid Fermentation! Less than **7%** of stored energy is extracted.

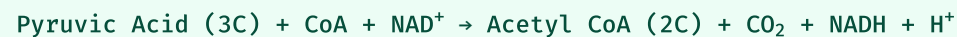
12.4 Aerobic Respiration & TCA Cycle

NCERT Pages 158–159

Link Reaction & Tricarboxylic Acid (Krebs) Cycle in Mitochondrial Matrix

THE LINK (GATEKEEPER) REACTION

Pyruvate enters the mitochondrial matrix from the cytoplasm and undergoes **Oxidative Decarboxylation** catalyzed by the *Pyruvate Dehydrogenase* multienzyme complex (requiring Mg²⁺, NAD⁺, Coenzyme A, TPP, and Lipoic acid):



🧠 KREBS CYCLE 8-STEP MNEMONIC:

"Can I Keep Selling Substances For Money, Officer?"

Citrate (6C) → Isocitrate (6C) → α-Ketoglutarate (5C) → Succinyl-CoA (4C) → Succinate (4C) → Fumarate (4C) → Malate (4C) → Oxaloacetate (4C)

TCA CYCLE ENERGETIC STOICHIOMETRY SUMMARY

NADH Produced
3 NADH + H⁺

FADH₂ Produced
1 FADH₂

Direct ATP (GTP)
1 ATP

Summary Equation per Pyruvic Acid:
Pyruvic acid + 4 NAD⁺ + FAD⁺ + 2 H₂O + ADP + P_i → 3 CO₂ + 4 NADH + 4 H⁺ + FADH₂ + ATP

12.4.2 Electron Transport System (ETS) & Complex V

NCERT Pages 159–161

Inner Mitochondrial Membrane (Cristae) • Oxidative Phosphorylation

TABLE 12.4: COMPREHENSIVE ARCHITECTURE OF ETS COMPLEXES I TO V

COMPLEX	NAME	PROSTHETIC GROUPS	ELECTRON PATHWAY & PROTON PUMPING
Complex I	NADH Dehydrogenase	FMN, Fe-S	Transfers e ⁻ from NADH to Ubiquinone (UQ). Pumps 4 H⁺ .
Complex II	Succinate Dehydrogenase	FAD, Fe-S	Transfers e ⁻ from FADH ₂ to Ubiquinone (UQ). Pumps 0 H ⁺ .
Complex III	Cytochrome bc ₁ Complex	Cyt b, Fe-S, Cyt c ₁	Receives e ⁻ from UQH ₂ → passes to Cytochrome c. Pumps 4 H⁺ .
Complex IV	Cytochrome c Oxidase	Cyt a, Cyt a ₃ , Cu _A , Cu _B	Passes e ⁻ to terminal O ₂ to form H ₂ O. Pumps 2 H⁺ .
Complex V	ATP Synthase	F ₀ Channel + F ₁ Headpiece	Proton flux down PMF from intermembrane space to matrix. 4 H⁺ = 1 ATP.

MOBILE ELECTRON CARRIERS:

- Ubiquinone (UQ / CoQ): Hydrophobic carrier within inner membrane; accepts e⁻ from Complex I & II.
- Cytochrome c: Small soluble protein on the *outer surface of inner membrane*; shuttles e⁻ between Complex III and Complex IV.

TERMINAL HYDROGEN ACCEPTOR:

Oxygen acts as the ultimate electron and proton sink at Complex IV, forming metabolic water:



12.5 The Respiratory Balance Sheet

NCERT Pages 161–162

Calculating net ATP yield per glucose oxidised relies on 4 Theoretical Assumptions:

1. Sequential Pathway

Glycolysis → Link Rxn → TCA → ETS operate in order.

2. NADH Transfer

Cytosolic NADH enters mitochondria for ETS.

3. No Intermediate Loss

Intermediates are not withdrawn for biosynthesis.

4. Glucose Substrate

Glucose is the sole respiratory substrate.

RESPIRATION STAGE	DIRECT ATP (SLP)	COENZYME OUTPUT	OXIDATIVE ATP VIA ETS	STAGE TOTAL
1. Glycolysis	2 ATP	2 NADH	2 × 3 = 6 ATP	8 ATP
2. Link Reaction	0 ATP	2 NADH	2 × 3 = 6 ATP	6 ATP
3. Krebs Cycle (2 Turns)	2 ATP (GTP)	6 NADH + 2 FADH ₂	(6 × 3) + (2 × 2) = 22 ATP	24 ATP
NET THEORETICAL ATP YIELD PER GLUCOSE				38 ATP

12.6 Amphibolic Pathway

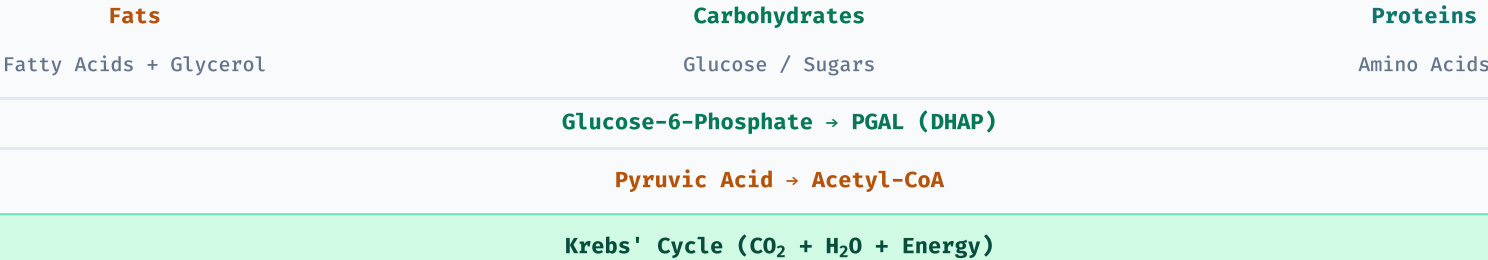
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Respiration involves both **Catabolism** (breakdown of organic molecules for ATP) and **Anabolism** (withdrawal of intermediates for biosynthesis). Thus, it is correctly designated as an **AMPHIBOLIC PATHWAY**.

SUBSTRATE ENTRY & WITHDRAWAL POINTS:

- **Fats:** Hydrolyzed into Fatty Acids (converted to Acetyl-CoA via β -oxidation) + Glycerol (converted to PGAL).
- **Proteins:** Hydrolyzed to Amino Acids; enter as Pyruvate, Acetyl-CoA, or Krebs intermediates (α -ketoglutarate, succinyl-CoA).
- **Anabolic Withdrawal:** Acetyl-CoA is withdrawn to synthesize fats; Krebs intermediates are withdrawn to synthesize proteins/pigments.

INTERRELATIONSHIP MATRIX



12.7 Respiratory Quotient (RQ)

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MATHEMATICAL DEFINITION

Respiratory Quotient (RQ) = (Volume of CO_2 Evolved) / (Volume of O_2 Consumed)

SUBSTRATE	STOICHIOMETRIC EQUATION	RQ VALUE	SIGNIFICANCE
Carbohydrates	$\text{C}_6\text{H}_{12}\text{O}_6 + 6 \text{ O}_2 \rightarrow 6 \text{ CO}_2 + 6 \text{ H}_2\text{O}$	1.0 (Unity)	Equal CO_2 evolved and O_2 consumed.
Fats (Tripalmitin)	$2 (\text{C}_{51}\text{H}_{98}\text{O}_6) + 145 \text{ O}_2 \rightarrow 102 \text{ CO}_2 + 98 \text{ H}_2\text{O}$	0.7 (< 1.0)	Oxygen-poor substrate; requires high O_2 input.
Proteins	Complex deamination breakdown	~0.9 (< 1.0)	Requires more O_2 relative to CO_2 output.

SUBSTRATE	STOICHIOMETRIC EQUATION	RQ VALUE	SIGNIFICANCE
Organic Acids	$C_4H_6O_5 + 3 O_2 \rightarrow 4 CO_2 + 3 H_2O$	1.33 (> 1.0)	Oxygen-rich substrate; requires less external O ₂ .
Anaerobic	Glucose → 2 Ethanol + 2 CO ₂ (No O ₂)	Infinity (∞)	Zero oxygen consumed.

? Complete NCERT Textbook Exercise Solutions

NCERT Pages 164–165

- Q1. Differentiate between: (a) Respiration and Combustion (b) Glycolysis and Krebs' cycle (c) Aerobic respiration and Fermentation ▼
- Q2. What are respiratory substrates? Name the most common respiratory substrate. ▼
- Q4, Q8, Q9, Q11 & Q12. High-Yield Conceptual Answers ▼